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ORIGINAL RESEARCH

Effectiveness of a Multicomponent Quality Improvement Strategy to Improve Achievement of Diabetes Care Goals

A Randomized, Controlled Trial

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Background: Achievement of diabetes care goals is suboptimal globally. Diabetes-focused quality improvement (QI) is effective but remains untested in South Asia.

Objective: To compare the effect of a multicomponent QI strategy versus usual care on cardiometabolic profiles in patients with poorly controlled diabetes.

Design: Parallel, open-label, pragmatic randomized, controlled trial. (ClinicalTrials.gov: NCT01212328)

Setting: Diabetes clinics in India and Pakistan.

Patients: 1146 patients (575 in the intervention group and 571 in the usual care group) with type 2 diabetes and poor cardiometabolic profiles (glycated hemoglobin [HbA_{1c}] level ≥8% plus systolic blood pressure [BP] ≥140 mm Hg and/or low-density lipoprotein cholesterol [LDLc] level ≥130 mg/dL).

Intervention: Multicomponent QI strategy comprising nonphysician care coordinators and decision-support electronic health records.

Measurements: Proportions achieving HbA_{1c} level less than 7% plus BP less than 130/80 mm Hg and/or LDLc level less than 100 mg/dL (primary outcome); mean risk factor reductions, health-related quality of life (HRQL), and treatment satisfaction (secondary outcomes).

Results: Baseline characteristics were similar between groups. Median diabetes duration was 7.0 years; 6.8% and 39.4% of participants had preexisting cardiovascular and microvascular disease, respectively; mean HbA_{1c} level was 9.9%; mean BP was 143.3/81.7 mm Hg; and mean LDLc level was 122.4 mg/dL. Over a median of 28 months, a greater percentage of intervention participants achieved the primary outcome (18.2% vs. 8.1%; relative risk, 2.24 [95% CI, 1.71 to 2.92]). Compared with usual care, intervention participants achieved larger reductions in HbA_{1c} level (−0.50% [CI, −0.69% to −0.32%]), systolic BP (−4.04 mm Hg [CI, −5.85 to −2.22 mm Hg]), diastolic BP (−2.03 mm Hg [CI, −3.00 to −1.05 mm Hg]), and LDLc level (−7.86 mg/dL [CI, −10.90 to −4.81 mg/dL]) and reported higher HRQL and treatment satisfaction.

Limitation: Findings were confined to urban specialist diabetes clinics.

Conclusion: Multicomponent QI improves achievement of diabetes care goals, even in resource-challenged clinics.

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Diabetes is one of the fastest-spreading, most burdensome, and most costly chronic diseases, affecting an estimated 415 million persons worldwide (1). Diabetes management focuses on reducing patients' risk for microvascular (retinopathy, nephropathy, and neuropathy) and macrovascular (coronary, cerebrovascular, and peripheral vascular diseases) complications through control of blood glucose levels (2–5), blood pressure (BP) (6–8), and lipid levels (9, 10) and avoidance of tobacco (11, 12). Although evidence-based guidelines (13, 14) strongly recommend comprehensively controlling these cardiometabolic parameters (15, 16), large gaps exist in the achievement of care goals in actual practice globally (17–21).

To address gaps that cannot be fully addressed with new drugs or devices, quality improvement (QI) interventions directed at patients (for example, reminders), providers (for example, guideline prompts), and health systems (for example, institutionalizing a "culture of quality") (22) can improve adherence, risk factor control, and patient satisfaction (23, 24). Indeed, QI may have contributed to encouraging data from the United States showing risk factor improvements and parallel reductions in diabetes complications over the past 2 decades (25, 26). However, gaps remain (19) and are even more substantial in low- and middle-income countries (LMICs) (20, 21) that have large and growing diabetes populations and fewer resources (1). This is especially true in South Asia, where barriers at the patient level (for example, low motivation), provider level (for example, clinical inertia), and system level (for example, fragmentation of care) result in small proportions of patients achieving diabetes care goals (21). In addition, most evidence comes from trials of single QI interventions (27). In reality, however, care gaps are due to multiple interacting fac-

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tors: Poor patient adherence and lack of treatment intensification by providers each explain roughly half of cases of poor disease control among patients with diabetes (28, 29).

We evaluated a multicomponent diabetes QI strategy to address patient and provider barriers in a pragmatic implementation trial in South Asia, a challenging LMIC region with more than 76 million persons affected by diabetes and the highest number of annual diabetes-related deaths worldwide (1). We tested whether multicomponent QI, compared with usual care, could improve achievement of diabetes care goals over 2.5 years among patients with multiple poorly controlled cardiometabolic parameters.

METHODS

Design Overview

The CARRS (Center for cArdio-metabolic Risk Reduction in South Asia) trial was a multicenter, prospective, parallel randomized, controlled, open-label pragmatic trial in which investigators were blinded to end points. Patients with poorly controlled type 2 diabetes attending 10 outpatient diabetes clinics in India and Pakistan were randomly assigned in a 1:1 ratio to a multicomponent care model (consisting of nonphysician care coordinators [CCs] and decision-support electronic health record [DS-EHR] software) or usual care. Detailed methods of the CARRS trial have been published separately (30).

The trial was registered at ClinicalTrials.gov (NCT01212328). Institutional ethics committees at each participating site and the coordinating centers (Public Health Foundation of India, New Delhi, India, and Emory University, Atlanta, Georgia) approved the study, and all eligible patients gave written informed consent before enrollment. Participants were enrolled from January 2011 to June 2012, and final follow-up visits were in July 2014.

Setting and Participants

CARRS trial sites were selected to include a diverse mix of publicly funded, semiprivate, and private outpatient clinics in India and Pakistan (**Appendix**, available at www.annals.org).

Men and women aged 35 years or older with type 2 diabetes and poor cardiometabolic profiles (glycated hemoglobin [HbA_{1c}] level ≥8% plus systolic BP ≥140 mm Hg and/or low-density lipoprotein cholesterol [LDLc] level ≥3.37 mmol/L [≥130 mg/dL]) who had attended the recruiting clinic for at least 3 months were eligible. We excluded persons with type 1 diabetes; rare forms of diabetes; or documented myocardial infarction, unstable angina, or stroke in the past 12 months.

Site investigators identified potentially eligible patients on the basis of elevated HbA_{1c} values and referred these patients to a screening officer (usually a clinic physician) for further baseline eligibility assessments. Subsequent annual assessments were conducted by a site physician. Because this was a pragmatic trial, site physicians were not purposefully blinded to the patient's intervention status and treated participants in both groups.

Randomization and Interventions

After baseline assessment, study staff at each clinic accessed each eligible participant's randomization allocation from a password-protected, Web-based data management system (Interactive Web Response System). The system randomly assigned participants in blocks of 4, and allocation was stratified by site.

The choice of QI components for the intervention was based on extensive literature searches, conversations among the investigators who designed the study, and conversations with clinic site investigators with regard to feasibility and acceptability. To our knowledge, the package of QI strategies that we used and its components have not been tested in South Asia.

The intervention as a whole focused on improving patient self-care and facilitating better monitoring and treatment intensification by providers. Usual care participants continued to be treated at the discretion of their clinic physicians.

Intervention participants were supported by nonphysician CCs in addition to their usual physicians. The CCs individualized follow-up based on patients' risk level and adherence. At a minimum, per the study protocol, the CC was responsible for following up with intervention patients at least once every 3 months to set up laboratory or clinic appointments and contacted patients by telephone at least once a month to discuss diabetes self-management, adherence to diet plans, exercise, tobacco cessation, medication use, self-monitoring of glucose levels (if the patient was receiving insulin), and stress management.

Treatments were aligned with evidence-based guidelines through individualized computer-generated clinical prompts; the DS-EHR integrated patients' consultation and laboratory data and used adapted algorithms based on prevailing clinical guidelines (31), with a preference for low-cost generic medications. The CCs had a distinct access level in the DS-EHR and recorded their interactions with patients. To encourage responsive and appropriate treatment intensification by physicians, CCs were responsible for entering updated patient indicators into the DS-EHR; they then printed management prompts generated by the DS-EHR and reviewed them with the treating physician immediately instead of waiting for a subsequent patient visit. Physicians could, at their discretion, accept or reject care prompts and modify management plans on the basis of clinical judgment as long as justification was documented in the DS-EHR.

With the goal of scalability in low-resource settings, selection criteria for CCs included nonphysicians with training in allied health fields (such as dietetics or social work), at least 6 months of health care experience, and good organizational and basic computing skills. The CCs received 2.5 days of intensive training before enrollment, with half-day refresher sessions at annual study meetings related to diabetes, barriers to treat-

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ment, supporting treatment changes, and motivational interviewing. The CCs also had 2 teleconference sessions with an experienced endocrinologist to discuss common barriers and strategies to overcome them.

Outcomes and Follow-up

The primary outcome was the proportion of patients from each group achieving an HbA_{1c} level less than 7% plus a BP less than 130/80 mm Hg and/or an LDLc level less than 2.59 mmol/L (<100 mg/dL) (<1.81 mmol/L [<70 mg/dL] for patients with a history of cardiovascular disease [CVD]). We also examined between-group differences in the following secondary outcomes: achievement of individual risk factor targets, mean changes in risk factors, and patient-reported outcomes (health-related quality of life [HRQL] and treatment satisfaction scores). Other secondary outcomes that we assessed included processes of care, medication use, self-care activities, acceptability of the intervention from the provider and patient perspectives, and cost-effectiveness; these will be reported elsewhere.

Participants in both groups attended baseline (before randomization), annual, and end-of-study (EOS) data collection visits. The EOS visit occurred between 24 and 36 months after randomization. Because this was a pragmatic trial focused on generalizability, data collection visits were paid for by the study, but costs of interim follow-up visits, testing, medications, or procedures recommended by physicians for either group were borne by the patients themselves (or in publicly funded settings, their clinics).

Annual data collection visits included HbA_{1c}, BP, and LDLc measurements. Blood pressure was measured using electronic devices (Omron T9P), and paper printouts were logged. Blood samples were analyzed by local laboratories that were enrolled in an external quality assurance scheme. Participants reported self-care activities (diet and exercise) and adherence to medications as the number of days the diet and exercise plan was followed or medication was taken, respectively, in the week before the visit (values ranging from 0 to 7 days). The research team also asked participants to rate their HRQL (using the EuroQol-5D visual analogue scale, with scores ranging from 0 to 100) and treatment satisfaction (using the Diabetes Treatment Satisfaction Questionnaire, with scores ranging from 0 to 32). Participants were asked open-ended questions about macrovascular and microvascular diabetes complications and any adverse events (for example, hypoglycemia), serious adverse events (for example, myocardial infarction), or hospitalizations. The severity (mild, moderate, or severe) of all reported adverse events was assessed using the Hartwig and Siegel scale (32). In addition, site investigators assessed the likelihood that serious adverse events were causally related to the study intervention by using the Uppsala monitoring center guidelines (33). Full study measures and definitions have been published in greater detail elsewhere (30).

Statistical Analysis

A priori sample size calculations showed that 1120 participants (560 in each group) would offer greater than 85% power (α, 0.05) to detect a 40% relative difference (28% vs. 20% in absolute terms) in achievement of the primary outcome, accounting for an expected 20% loss to follow-up.

All analyses were based on intention-to-treat principles. A 2-sided *P* value less than 0.05 was used to indicate statistical significance. We used Stata, version 14.0 (StataCorp), for all analyses.

We compared baseline participant characteristics in the intervention and usual care groups by using the *t* test or Wilcoxon rank-sum test (for normally distributed and skewed continuous variables, respectively) and the chi-square test or Fisher exact test (for categorical variables with adequate and sparse cell counts, respectively). For the primary analysis, to determine the relative risks (RRs) for achievement of multiple risk factor targets using all longitudinal time points available, we estimated log-binomial models with a generalized estimating equation approach to account for correlation of observations within participants over time. Mean differences in continuously specified HbA_{1c} level, systolic and diastolic BPs, and LDLc level were also determined using linear regression with a generalized estimating equation approach. Both sets of models were adjusted for baseline values, treatment group, time, treatment-by-time interaction, and site.

We examined for heterogeneity of effect across sites and by age (35 to 49, 50 to 64, and ≥65 years), sex, education (up to primary school, secondary school, or college graduate or above), monthly income (<$200, $200 to $400, or >$400), public or private clinic, body mass index (BMI) (<25, 25 to 29.9, or ≥30 kg/m^2), duration of diabetes (<7 or ≥7 years, based on median duration of follow-up), history of CVD, history of microvascular complications, insulin use, baseline HbA_{1c} level (≥9% vs. <9%), baseline systolic BP (≥140 vs. <140 mm Hg), and baseline LDLc level (≥130 vs. <130 mg/dL). For each of the aforementioned subgroups, we repeated the primary analysis with the addition of the subgroup variable along with its interaction with treatment. Heterogeneity was assessed based on the significance of the interaction term.

We estimated within-group changes between baseline and the end of the study in HbA_{1c} level, BP, LDLc level, BMI, weight, waist circumference, creatinine level, albumin–creatinine ratio, HRQL, and treatment satisfaction scores by using paired *t* tests and Wilcoxon signed-rank tests. We estimated mean differences by treatment group in baseline-to-EOS changes using analysis of covariance, with adjustment for baseline values.

We assessed the mechanism of missingness by using the Little test (34). We also conducted additional analyses to examine the sensitivity of our primary results to missing outcome data by applying 2 methods: inverse probability weighting (35) and multiple imputation using chained equations (36). The former approach weighted the analysis by the inverse of the pre-

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dicted probability of being observed at any given visit. The predicted probability of being observed was computed based on a logistic model with treatment group, site, and all baseline risk factors as predictors. In the latter approach we used a fully conditional specification method to impute missing outcome data. We used a linear regression model that contained treatment group; age; sex; education; site; and baseline values for height, weight, waist circumference, plasma glucose level, HbA_{1c} level, systolic and diastolic BP, triglyceride level, LDLc level, and high-density lipoprotein cholesterol level. We also assessed convergence by plotting means and SDs by iteration and imputation. We conducted 10 imputations and analyzed the results according to convention (37).

Role of the Funding Source

This study was funded by the National Heart, Lung, and Blood Institute of the National Institutes of Health and by UnitedHealth Group. The funding sources were not involved in designing the study, collecting or interpreting the data, or writing the manuscript.

RESULTS

Of 1486 participants who were screened, 1146 (575 in the intervention group and 571 in the usual care group) were eligible and were randomly assigned. Median follow-up was 28 months (range, 22 to 36 months), and 1027 participants (516 in the intervention group and 511 in the usual care group) completed EOS visits (89.6% retention). Of these, 467 intervention participants and 463 usual care participants attended the clinic for full questionnaire, clinical, and biochemical assessments, and 49 and 48, respectively, completed the EOS visit at home or by telephone (**Appendix Figure 1**; available at www.annals.org).

At enrollment, participant characteristics between study groups were similar (**Table**), except a higher proportion of intervention group participants were insulin users (46.3% vs. 39.8%; $P = 0.026$). Participants' mean age was 54.2 years, 45.9% were male, 70.1% had some high school education, and 33.6% were considered high-income (monthly income >$400). At baseline, mean HbA_{1c} level was 9.9%, mean BP was 143.3/81.7 mm Hg, and mean LDLc level was 3.17 mmol/L (122.4 mg/dL). Most participants (70.9%) had an HbA_{1c} level of at least 8% and a systolic BP of at least 140 mm Hg; 47.1% had an HbA_{1c} level of at least 8% and an LDLc level of at least 130 mg/dL; and 18.1% had an HbA_{1c} level of at least 8%, a systolic BP of at least 140 mm Hg, and an LDLc level of at least 130 mg/dL. Mean BMI was 27.5 kg/m^2, mean waist circumference was 96.1 cm, and 2.8% of participants were current smokers. Median duration of diabetes was 7.0 years, 6.8% of participants had preexisting CVD, and 39.4% had previous microvascular complications.

Six CCs had undergraduate degrees in life sciences, 2 were dieticians, 1 was trained in computer applications, and 1 was a nurse. Across sites, CCs managed 50 to 80 intervention group patients each. Their primary responsibilities were to motivate diabetes self-management among patients, facilitate follow-up and preventive screening visits, and encourage proactive follow-up and treatment modifications by physicians with the assistance of the DS-EHR. The CCs individualized their follow-up frequency and interactions with patients on the basis of risk level and motivation. In addition, sites varied in how intensively they implemented the intervention. A detailed process evaluation across and within sites was outside the scope of this analysis.

During the study, more than twice as many intervention participants achieved the primary outcome compared with usual care (18.2% vs. 8.1%; RR, 2.24 [95% CI, 1.71 to 2.92]) (**Figure 1**). Higher proportions of intervention participants achieved an HbA_{1c} level less than 7% (21.5% vs. 11.1%; RR, 1.93 [CI, 1.52 to 2.45]), a BP less than 130/80 mm Hg (51.0% vs. 45.0%; RR, 1.14 [CI, 1.04 to 1.26]), and an LDLc level less than 100 mg/dL (<70 mg/dL for those with previous CVD) (56.4% vs. 47.1%; RR, 1.23 [CI, 1.13 to 1.34]). Among participants with a history of CVD (6.5%), we found no significant between-group difference in achievement of an LDLc level less than 70 mg/dL.

Intervention participants achieved greater reductions in HbA_{1c} level (−0.50% [CI, −0.69% to −0.32%]), systolic BP (−4.04 mm Hg [CI, −5.85 to −2.22 mm Hg]), diastolic BP (−2.03 mm Hg [CI, −3.00 to −1.05 mm Hg]), and LDLc level (−0.20 mmol/L [−7.86 mg/dL] [CI, −0.28 to −0.12 mmol/L {−10.90 to −4.81 mg/dL}]) (**Appendix Figure 2**, available at www.annals.org). They also reported greater baseline-to-EOS improvements in HRQL and treatment satisfaction scores, but not weight, BMI, waist circumference, or albumin–creatinine ratio (**Appendix Table 1**, available at www.annals.org).

In sensitivity analyses to address missing data (**Appendix Table 2**, available at www.annals.org), we noted no relative or absolute differences in the between-group likelihood of achieving the primary outcome. Achieving the primary outcome did not statistically significantly differ across sites (**Appendix Table 3**, available at www.annals.org) or by age, sex, baseline metabolic profile, duration of diabetes, insulin use, or public or private clinic setting (**Figure 2**). However, compared with participants with education up to primary school and the lowest-income earners, college-educated or high-income participants were 2 to 3 times more likely to achieve the primary outcome ($P = 0.006$ and 0.007, respectively).

With regard to serious adverse events, we found no statistically significant between-group differences in hypoglycemic episodes requiring hospitalization (4 vs. 1; $P = 0.21$), new microvascular complications (163 vs. 190; $P = 0.080$), or new macrovascular events (33 vs. 35; $P = 0.77$), although intervention participants reported more mild hypoglycemic episodes (133 vs. 21; $P < 0.001$) (**Appendix Table 4**, available at www.annals.org).

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Table. Baseline Characteristics, by Treatment Group*

Characteristic	Intervention	Usual Care
Participants, *n*	575	571
Mean age (SD), *y*	54.2 (9.2)	54.2 (9.2)
Male, %	44.9	47.1
Education, %		
Up to primary school	29.4	29.4
Secondary school	44.9	42.2
College graduate or above	25.0	27.9
Unknown	0.7	0.5
Monthly income, %		
<10 000 INR (<200 USD)	35.5	39.9
10 000–20 000 INR (200–400 USD)	17.6	18.4
>20 000 INR (>400 USD)	35.3	31.9
Unknown	11.7	9.8
Median duration of diabetes (IQR), *y*	7 (3–13)	7 (3–12)
Current smoker, %†	2.4	3.5
Comorbidities, %‡		
Previous cardiovascular disease§	7.0	6.7
Myocardial infarction	3.1	3.0
Coronary heart disease	6.9	6.7
Stroke	2.3	1.8
Previous peripheral vascular disease‖	6.1	5.4
Microvascular complications	39.0	39.8
Retinopathy	8.5	8.9
Neuropathy¶	33.9	33.4
Nephropathy	9.0	10.5
Depression	2.1	2.3
Mean examination findings (SD)		
Waist circumference, *cm*	96.1 (11.8)	96.1 (10.9)
Weight, *kg*	69.4 (12.6)	68.9 (13.3)
Body mass index, *kg/m*^{2}	27.5 (4.6)	27.4 (5.4)
Heart rate, *beats/min*	84.4 (11.9)	83.4 (13.0)
Metabolic variables		
Mean hemoglobin A_{1c} level (SD), %	9.9 (1.5)	9.9 (1.7)
Mean fasting blood glucose level (SD)		
mmol/L	9.93 (3.43)	9.79 (3.68)
mg/dL	178.9 (61.8)	176.4 (66.3)
Mean low-density lipoprotein cholesterol level (SD)		
mmol/L	3.15 (0.93)	3.19 (0.98)
mg/dL	121.5 (36.1)	123.2 (37.7)
Mean high-density lipoprotein cholesterol level (SD)		
mmol/L	1.14 (0.23)	1.14 (0.24)
mg/dL	44.1 (8.7)	43.9 (9.1)
Median triglyceride level (IQR)		
mmol/L	1.56 (1.21–2.11)	1.60 (1.18–2.24)
mg/dL	138.0 (107.0–187.0)	142.0 (104.0–198.0)
Mean total cholesterol level (SD)		
mmol/L	5.04 (1.17)	5.07 (1.15)
mg/dL	194.7 (45.3)	195.7 (44.3)
Mean systolic blood pressure, *mm Hg*	144.2 (18.8)	142.4 (20.0)
Mean diastolic blood pressure, *mm Hg*	82.3 (11.0)	81.0 (10.8)
Mean creatinine level (SD)		
μmol/L	83.1 (35.4)	89.3 (79.6)
mg/dL	0.94 (0.40)	1.01 (0.90)
Median urinary albumin–creatinine ratio (IQR), *mg/g*	7.60 (0.16–58.00)	8.10 (0.12–63.00)
Medication use, %		
Oral hypoglycemic agents	95.0	94.1
Insulin	46.3	39.8
Oral hypoglycemic agents and insulin	28.4	25.4
Blood pressure–lowering drugs	64.2	60.4
Lipid-lowering drugs	59.1	59.0
Mean EuroQol-5D score (SD)	69.8 (20.1)	70.2 (19.3)
Mean Diabetes Treatment Satisfaction Questionnaire score (SD)	28.4 (6.4)	28.1 (6.7)

INR = Indian rupees; IQR = interquartile range; USD = U.S. dollars.
* All baseline characteristics were similar between treatment groups except those for insulin use. Percentages may not sum to 100 due to rounding.
† Refers to cigarette, beedi, and cigar smoking (self-reported; participants were asked about current regular use of these products).
‡ Self-reported by patients and confirmed by a physician.
§ Includes acute coronary syndrome (myocardial infarction or unstable angina), angioplasty (percutaneous coronary intervention) or open heart surgery (coronary artery bypass graft surgery), chronic stable angina, ischemic or hemorrhagic stroke, transient ischemic attack, arrhythmia requiring medical intervention, or heart failure.
‖ Includes intermittent claudication or rest pain in legs.
¶ Includes loss of sensation in hands/feet or sexual complications (erectile dysfunction).

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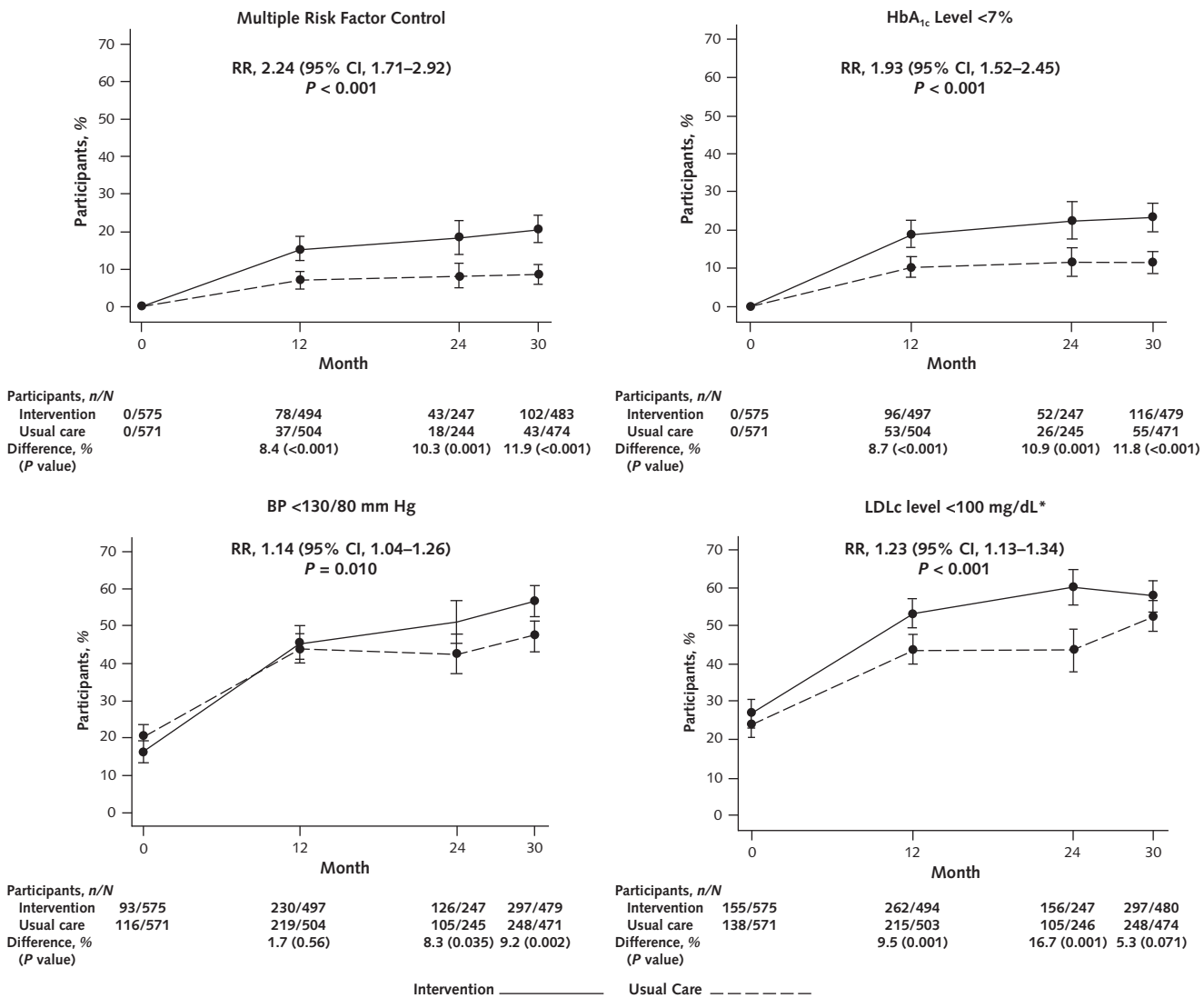
Figure 1. Control of risk factors during follow-up, by treatment group.



Overall RR was obtained via log-binomial models using generalized estimating equations. Estimates combined all nonmissing values collected at baseline, months 12 and 24, and the end of the study. Error bars indicate 95% CIs. Model terms included treatment, time, treatment-by-time interaction, baseline value, and site. BP = blood pressure; HbA_{1c} = glycated hemoglobin; LDLc = low-density lipoprotein cholesterol; RR = relative risk.
* <70 mg/dL for participants with history of cardiovascular disease.

DISCUSSION

In this multicenter trial among patients with diabetes and poor cardiometabolic profiles in India and Pakistan, a multicomponent QI strategy was associated with twice the probability of achieving combined diabetes care goals and larger baseline-to-EOS reductions for each risk factor compared with usual care. Similar effects were observed in public and private clinics, and the intervention was not associated with excess serious adverse events. This study's contribution to the literature is that the intervention purposefully addressed multiple interacting barriers to care and was implemented in diverse and resource-challenged health care settings. More important, the diabetes QI intervention was associated with sustained health benefits among patients 2.5 years after randomization. Furthermore, unlike trials in which sponsors provide medications, visit costs, and self-care accessories, CARRS participants were responsible for all of their treatment and follow-up costs (except for annual study visits) and did not receive any additional incentives for their participation, which increases the generalizability of our findings. Given recent population-level improvements in cardiometabolic profiles of persons with diabetes and lower incidence rates of complications in the United States (25, 26) that were likely related to QI initiatives, these data suggest that such positive trends may be replicable in LMICs. If scaled, this may have implications for population health and health care costs in these countries.

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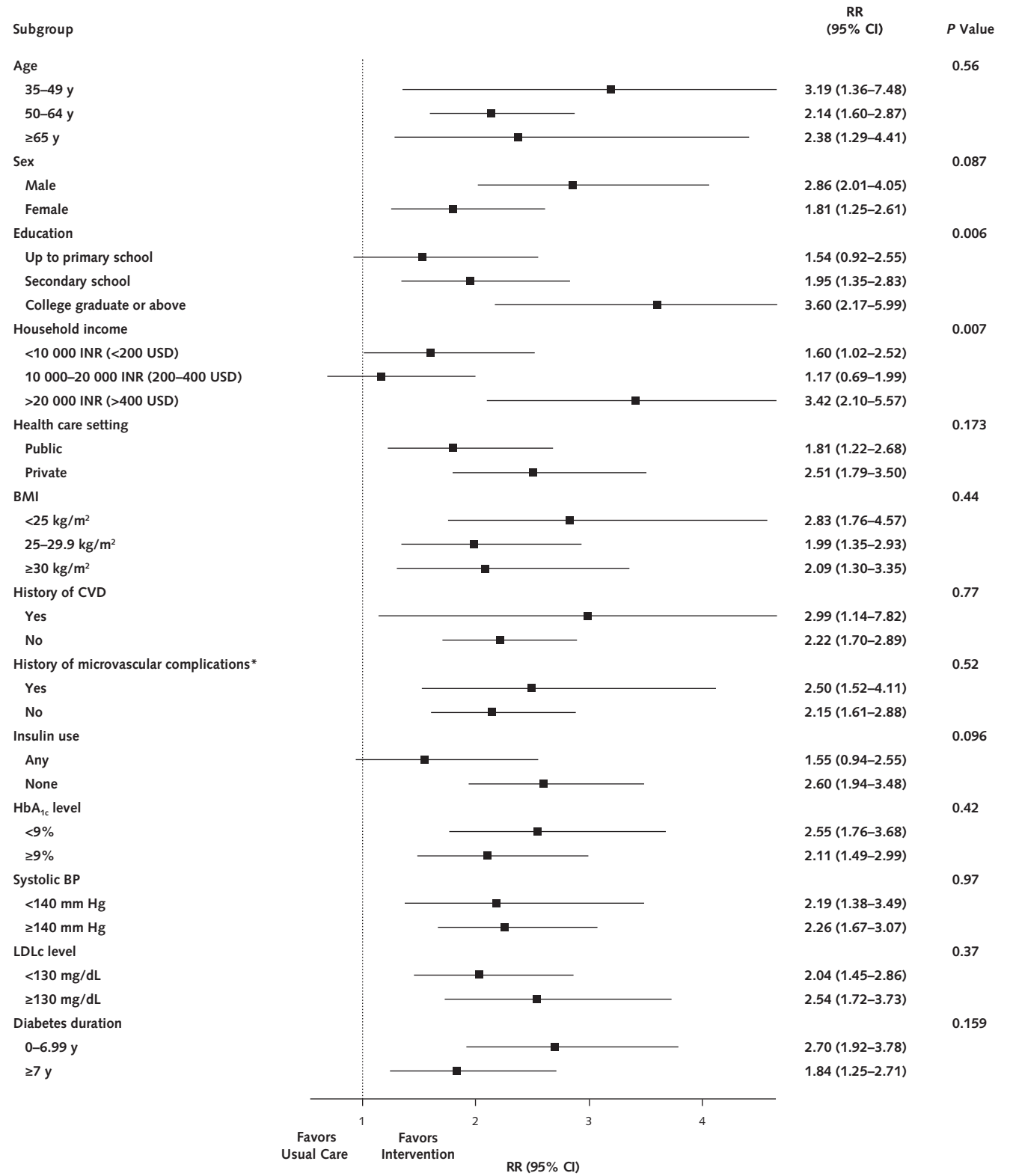
Figure 2. Primary outcome (achievement of multiple risk factor targets), by prespecified baseline subgroups.



Error bars indicate 95% CIs, and *P* values are for the test of homogeneity for each subgroup. BMI = body mass index; BP = blood pressure; CVD = cardiovascular disease; HbA_{1c} = glycated hemoglobin; INR = Indian rupees; LDLc = low-density lipoprotein cholesterol; RR = relative risk; USD = U.S. dollars.
* Retinopathy, neuropathy, or renal failure.

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Few trials have evaluated comprehensive cardiometabolic improvements in patients with diabetes (15, 38, 39). The Steno-2 Study (based in Denmark, which has universal health care) enrolled 160 patients (mean age, 55 years) with diabetes and preexisting microalbuminuria, 25% of whom also had preexisting CVD. Over 8 years, the intervention group achieved greater reductions in HbA_{1c} level (−0.7%), BP (−10/−5 mm Hg), and LDLc level (−0.88 mmol/L [−34 mg/dL]), leading to reductions in microvascular and macrovascular disease and mortality of 50% to 70% (15, 16). Compared with Steno-2 (baseline HbA_{1c} level of about 8.6%, BP of about 148/86 mm Hg, and LDLc level of about 3.50 mmol/L [135 mg/dL]), CARRS trial participants had similar baseline characteristics, and both treatment groups experienced large within-group pre–post changes in HbA_{1c} level (−1.6% vs. −1.2%), BP (−18.3/−7.9 vs. −12.9/−4.9 mm Hg), and LDLc level (−0.74 vs. −0.55 mmol/L [−28.5 vs. −21.3 mg/dL]) (**Appendix Table 1**). The sizeable spillover effects we observed in usual care participants reflect the real-life nature of QI and practice change, with physicians treating patients in both groups; therefore, our reported between-group effect size was conservative.

ADDITION (Anglo-Danish-Dutch Study of Intensive Treatment in People with Screen-Detected Diabetes in Primary Care) offers further context for our findings. In that study, clinics were randomly assigned to a provider-focused intervention (educating and auditing clinic staff to better manage glycemia, BP, and lipid levels) or control (40). Among 3057 persons with a mean age of 60 years, 9% of whom had previous CVD and a lower-risk baseline profile (HbA_{1c} level of approximately 7.0%, BP of approximately 149/86 mm Hg, and LDLc level of approximately 3.50 mmol/L [135 mg/dL]), both groups achieved within-group improvements, with slightly larger reductions in mean HbA_{1c} level (−0.08%), BP (−2.9/−1.4 mm Hg), and LDLc level (−0.20 mmol/L [−7.7 mg/dL]) among intervention participants. These between-group differences were associated with more modest (5% to 17%) and statistically nonsignificant reductions in cardiovascular and microvascular outcomes over 5 years (38, 41).

Our between-group findings fall between the Steno-2 and ADDITION findings. Whether and by how much the CARRS patient- and provider-focused intervention and related metabolic improvements will translate into future microvascular or macrovascular health benefits has yet to be seen and would be an informative contribution. The higher incidence of mild hypoglycemic episodes observed in CARRS participants to date reflects target-driven management; this undesirable effect may be mitigated through shared decision making between patients and providers, an aspect that is increasingly integrated into QI strategies.

Our findings also offer optimism for sustained effectiveness. In a meta-analysis of 48 cluster and 94 individual randomized, controlled trials testing diabetes QI interventions, the longest follow-up was just 12 months and the largest individual-focused trial included only 206 participants (27). The aggregate cardiometabolic improvements reported in this meta-analysis of studies largely from high-income settings (between-group differences of −0.37% in HbA_{1c} level, −3.1/−1.6 mm Hg in BP, and −0.10 mmol/L [−3.9 mg/dL] in LDLc level) were smaller than in our larger sample from low-resource countries at 2.5 years of follow-up (27).

In terms of acceptability, it was encouraging that intervention participants showed improvements in HRQL and treatment satisfaction, which suggests that they did not mind the intensity and extra follow-up of the intervention. Given that both groups had 4- to 6-point increases in overall well-being and 0.5- to 2.0-point increases in treatment satisfaction, the patients may have appreciated the extra attention provided by the intervention. It is hard to judge the clinical significance of these increases in patient-reported outcome scores because they were not anchored to clinical outcomes or stratified by achievement of care goals (42), but further investigation may provide additional insight. Finally, the patient-reported outcome scores reported are "global" average scores by group and may overshadow or hide specific physical, mental, or emotional domain changes that were more prominent.

Several limitations should be acknowledged. First, we were unable to examine the relative effectiveness of individual intervention components. However, the components included in our QI intervention are proven and seem to yield modest benefits on their own (23, 24, 27). Second, although the intervention had a decisive effect in doubling achievement of the primary outcome, in absolute terms, this amounted to only 20% and 50% of intervention participants achieving combined and individual care goals, respectively. Although much improvement is possible, achieving generic dichotomous targets (such as an HbA_{1c} level <7%) is increasingly less relevant in contemporary practice (43); instead, individualized, patient-centered goals are strongly emphasized (44, 45). Third, our findings were confined to specialist diabetes clinics in urban South Asia, and whether similar effects would be observed in primary care or rural settings remains to be seen. Finally, although contamination or spillover may have limited the observed between-group differences, this pragmatic trial aimed to replicate real-life settings. Thus, our effect size estimates are conservative, and these findings should be considered a proof of concept for other low-resource settings.

Diabetes prevalence and related morbidity and mortality have progressively escalated worldwide in the past 3 decades (1, 46), especially in LMICs. Our findings offer an encouraging demonstration of the implementation of comprehensive diabetes management and QI in LMIC settings. Broader implementation of QI strategies like those used in CARRS offers the potential for achievement of diabetes care goals and preservation of quality of life for millions.

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Appendix: CARRS Trial Group

Unless otherwise noted, the group members contributed to the manuscript but did not author it.

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* Authored the manuscript.

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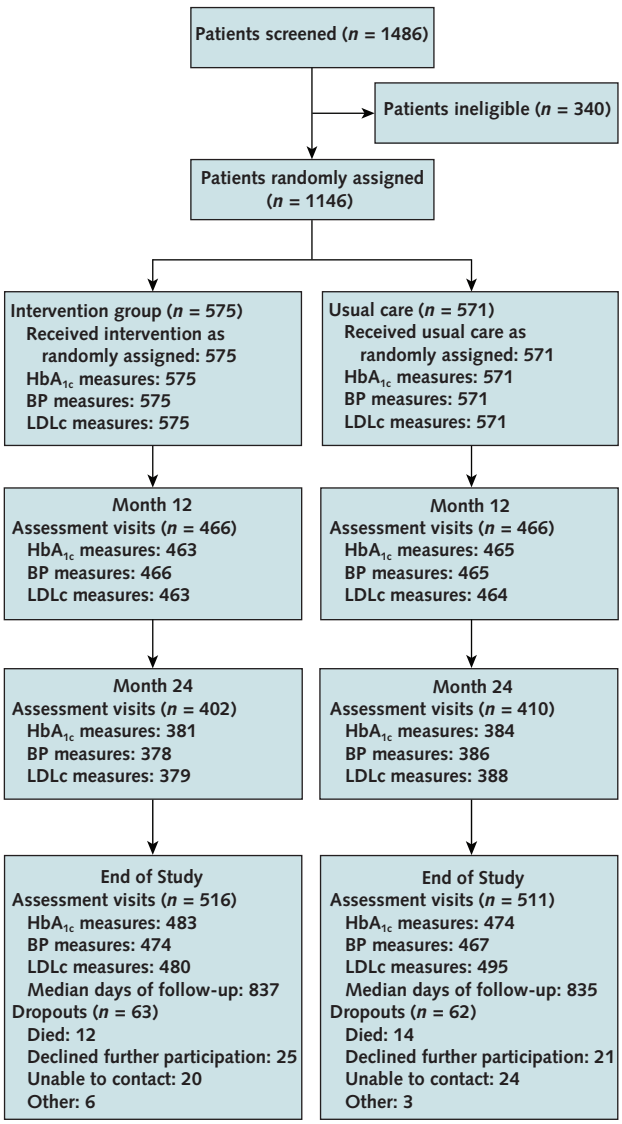
Appendix Figure 1. Registration, randomization, and follow-up of study participants.



BP = blood pressure; HbA_{1c} = glycated hemoglobin; LDLc = low-density lipoprotein cholesterol.

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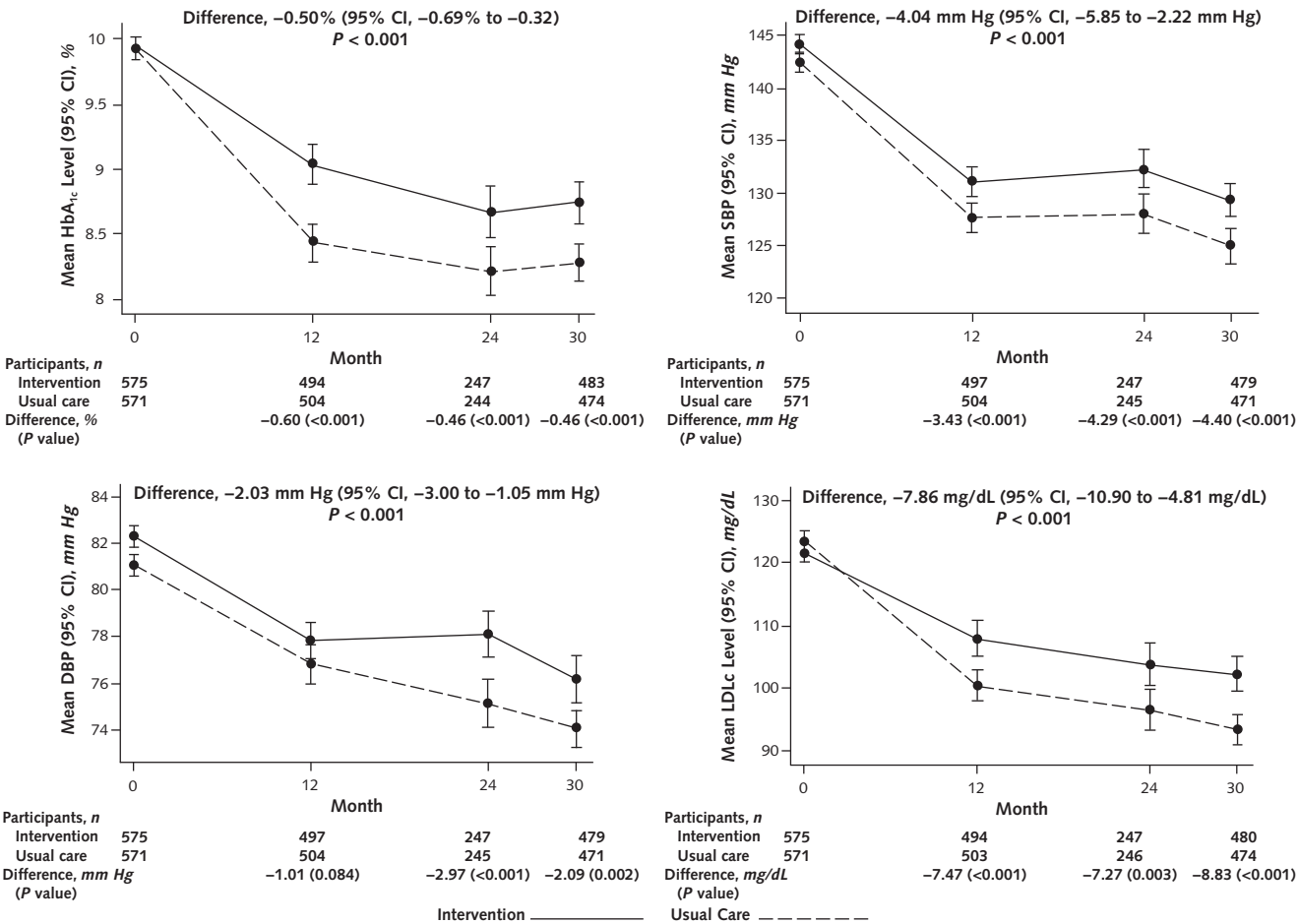
Appendix Figure 2. Mean between-group changes in HbA_{1c} level, SBP, DBP, and LDLc level during follow-up.



Overall mean differences were obtained via linear regression models using generalized estimating equations. Estimates combine all nonmissing values collected at baseline, months 12 and 24, and the end of the study. Error bars indicate 95% CIs. Model terms included treatment, time, treatment-by-time interaction, baseline value, and site. DBP = diastolic blood pressure; HbA_{1c} = glycated hemoglobin; LDLc = low-density lipoprotein cholesterol; SBP = systolic blood pressure.

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Appendix Table 1. Mean Changes in Metabolic and Patient-Reported Outcomes During the Study and Differences Between Treatment Groups*

Variable	Mean Change Between End of Study and Baseline (95% CI)		Mean Difference Between Groups (95% CI)†	*P* Value
	Intervention	Usual Care		
Hemoglobin A_{1c} level, *%*	−1.6 (−1.8 to −1.4)	−1.2 (−1.4 to −0.98)	−0.4 (−0.7 to −0.19)	<0.001
Systolic blood pressure, *mm Hg*	−18.3 (−20.4 to −16.2)	−12.9 (−14.7 to −11.0)	−5.4 (−8.2 to −2.6)	<0.001
Diastolic blood pressure, *mm Hg*	−7.9 (−9.0 to −6.8)	−4.9 (−6.2 to −3.7)	−2.9 (−4.6 to −1.3)	<0.001
Low-density lipoprotein cholesterol level, *mg/dL*	−28.5 (−31.8 to −25.2)	−21.3 (−25.1 to −17.6)	−7.1 (−12.1 to −2.1)	0.005
Weight, *kg*	0.45 (−0.04 to 0.94)	0.35 (−0.15 to −0.86)	−0.09 (−0.81 to 0.62)	0.79
Body mass index, *kg/m²*	0.21 (0.01 to 0.41)	0.16 (−0.04 to 0.37)	−0.05 (−0.33 to 0.23)	0.73
Waist circumference, *cm*	−0.23 (−0.99 to 0.54)	0.34 (−0.51 to 1.20)	0.56 (−0.58 to 1.71)	0.33
EuroQol-5D score	5.9 (4.4 to 7.4)	4.1 (2.7 to 5.5)	1.8 (−0.2 to 3.9)	<0.001
Diabetes Treatment Satisfaction Questionnaire score	2.10 (1.59 to 2.69)	0.54 (0.15 to 1.20)	1.6 (0.7 to 2.5)	0.08
Creatinine level, *mg/dL*	0.087 (0.041 to 0.133)	0.014 (−0.064 to 0.092)	0.07 (−0.01 to 0.16)	0.11
Albumin–creatinine ratio, *mg/g*‡	0.49 (0.41 to 0.59)	0.49 (0.41 to 0.59)	–	0.35

* Paired *t* test used for within-group difference in mean values of all variables (end-of-study minus baseline values).
† Intervention minus usual care.
‡ Geometric mean is presented for log-transformed albumin–creatinine ratio, and *P* value for difference was calculated using the Wilcoxon signed-rank test.

Appendix Table 2. Sensitivity Analysis Using Inverse Probability Weighting and Multiple Imputations to Account for Missing Data

Variable	Relative Risk (95% CI)		
	Generalized Estimating Equation	Inverse Probability Weighting*	Multiple Imputation†
Combined risk factor control (hemoglobin A_{1c} level <7% plus blood pressure <130/80 mm Hg and/or low-density lipoprotein cholesterol level <100 mg/dL [<70 mg/dL for patients with history of cardiovascular disease])	2.24 (1.71–2.92)	2.23 (1.71–2.90)	2.11 (1.64–2.72)
Hemoglobin A_{1c} level <7%	1.93 (1.52–2.45)	1.91 (1.50–2.43)	1.89 (1.50–2.40)
Blood pressure <130/80 mm Hg	1.14 (1.04–1.26)	1.15 (1.03–1.26)	1.18 (1.01–1.30)
Low-density lipoprotein cholesterol level <100 mg/dL (<70 mg/dL for patients with history of cardiovascular disease)	1.23 (1.13–1.34)	1.23 (1.13–1.34)	1.22 (1.11–1.34)

* Estimates obtained by inverse weighting for missing outcomes using the generalized estimating equation.
† Estimates obtained by imputing the missing outcomes using fully conditional specification method and then using the generalized estimating equation.

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Appendix Table 3. Sitewise Achievement of Primary Outcomes, by Treatment Group

Primary Outcome	Participants, *n*					Relative Risk (95% CI)*
	Baseline	12 mo	24 mo	End of Study	Total	
Site 1						
Combined risk factor control (HbA_{1c} level <7% plus BP <130/80 mm Hg and/or LDLc level <100 mg/dL [<70 mg/dL for patients with a history of CVD])						
Intervention						
Controlled	0	3	6	11	20	4.71 (1.45–15.22)
Uncontrolled	75	63	40	53	231	–
Total	75	66	46	64	251	–
Usual care						
Controlled	0	1	1	4	6	–
Uncontrolled	75	60	51	61	247	–
Total	75	61	52	65	253	–
HbA_{1c} level <7%						
Intervention						
Controlled	0	6	6	14	26	1.64 (0.74–3.68)
Uncontrolled	75	60	40	51	226	–
Total	75	66	46	65	252	–
Usual care						
Controlled	0	6	3	6	15	–
Uncontrolled	75	55	49	61	240	–
Total	75	61	52	67	255	–
BP <130/80 mm Hg						
Intervention						
Controlled	16	24	14	19	73	0.71 (0.41–1.24)
Uncontrolled	59	42	32	46	179	–
Total	75	66	46	65	252	–
Usual care						
Controlled	19	30	17	26	92	–
Uncontrolled	56	31	35	37	159	–
Total	75	61	52	63	251	–
LDLc level <100 mg/dL (<70 mg/dL for patients with a history of CVD)						
Intervention						
Controlled	9	20	23	17	69	1.53 (1.02–2.29)
Uncontrolled	66	46	23	47	182	–
Total	75	66	46	64	251	–
Usual care						
Controlled	10	14	12	14	50	–
Uncontrolled	65	47	40	53	205	–
Total	75	61	52	67	255	–
Site 2						
Combined risk factor control (HbA_{1c} level <7% plus BP <130/80 mm Hg and/or LDLc level <100 mg/dL [<70 mg/dL for patients with a history of CVD])						
Intervention						
Controlled	0	9	4	7	20	0.99 (0.47–2.07)
Uncontrolled	81	60	37	57	235	–
Total	81	69	41	64	255	–
Usual care						
Controlled	0	12	5	3	20	–
Uncontrolled	80	54	28	56	218	–
Total	80	66	33	59	238	–
HbA_{1c} level <7%						
Intervention						
Controlled	0	12	5	8	25	0.98 (0.49–1.96)
Uncontrolled	81	57	36	56	230	–
Total	81	69	41	64	255	–
Usual care						
Controlled	0	14	5	5	24	–
Uncontrolled	80	52	28	54	214	–
Total	80	66	33	59	238	–
BP <130/80 mm Hg						
Intervention						
Controlled	13	25	24	43	105	1.59 (0.92–2.78)
Uncontrolled	68	44	17	21	150	–
Total	81	69	41	64	255	–

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Appendix Table 3–Continued

Primary Outcome	Participants, *n*					Relative Risk (95% CI)*
	Baseline	12 mo	24 mo	End of Study	Total	
Usual care						
Controlled	20	28	19	28	95	–
Uncontrolled	60	38	14	31	143	–
Total	80	66	33	59	238	–
LDLc level <100 mg/dL (<70 mg/dL for patients with a history of CVD)						
Intervention						
Controlled	27	40	29	46	142	1.35 (1.06–1.73)
Uncontrolled	54	29	12	18	113	–
Total	81	69	41	64	255	–
Usual care						
Controlled	20	32	14	24	90	–
Uncontrolled	60	34	19	35	148	–
Total	80	66	33	59	238	–
Site 3						
Combined risk factor control (HbA_{1c} level <7% plus BP <130/80 mm Hg and/or LDLc level <100 mg/dL [<70 mg/dL for patients with a history of CVD])						
Intervention						
Controlled	0	1	0	0	1	NA
Uncontrolled	5	4	4	5	18	–
Total	5	5	4	5	19	–
Usual care						
Controlled	5	3	3	2	13	–
Uncontrolled	5	3	3	2	13	–
Total	–	–	–	–	–	–
HbA_{1c} level <7%						
Intervention						
Controlled	0	1	1	0	2	NA
Uncontrolled	5	4	3	5	17	–
Total	5	5	4	5	19	–
Usual care						
Controlled	5	3	3	2	13	–
Uncontrolled	5	3	3	2	13	–
Total	–	–	–	–	–	–
BP <130/80 mm Hg						
Intervention						
Controlled	0	1	0	2	3	NA
Uncontrolled	5	4	4	3	16	–
Total	5	5	4	5	19	–
Usual care						
Controlled	2	1	3	3	9	–
Uncontrolled	3	2	0	0	5	–
Total	5	3	3	3	14	–
LDLc level <100 mg/dL (<70 mg/dL for patients with a history of CVD)						
Intervention						
Controlled	2	2	2	4	10	NA
Uncontrolled	3	3	2	1	9	–
Total	5	5	4	5	19	–
Usual care						
Uncontrolled	4	1	0	1	6	–
Controlled	1	2	3	1	7	–
Total	5	3	3	2	13	–
Site 4						
Combined risk factor control (HbA_{1c} level <7% plus BP <130/80 mm Hg and/or LDLc level <100 mg/dL [<70 mg/dL for patients with a history of CVD])						
Intervention						
Controlled	0	3	3	11	17	1.29 (0.52–3.20)
Uncontrolled	79	48	28	41	196	–
Total	79	51	31	52	213	–
Usual care						
Controlled	0	3	1	14	18	–
Uncontrolled	76	55	30	33	194	–
Total	76	58	31	47	212	–

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Appendix Table 3–Continued

Primary Outcome	Participants, *n*					Relative Risk (95% CI)*
	Baseline	12 mo	24 mo	End of Study	Total	
HbA_{1c} level <7%						
Intervention						
Controlled	0	4	4	13	21	1.04 (0.52–2.12)
Uncontrolled	79	47	27	41	194	–
Total	79	51	31	54	215	–
Usual care						
Controlled	0	6	2	14	22	–
Uncontrolled	76	52	29	33	190	–
Total	76	58	31	47	212	–
BP <130/80 mm Hg						
Intervention						
Controlled	17	29	21	40	107	1.07 (0.87–1.31)
Uncontrolled	62	22	10	12	106	–
Total	79	51	31	52	213	–
Usual care						
Controlled	16	34	15	37	102	–
Uncontrolled	60	24	16	10	110	–
Total	76	58	31	47	212	–
LDLc level <100 mg/dL (<70 mg/dL for patients with a history of CVD)						
Intervention						
Controlled	23	24	16	31	94	1.08 (0.83–1.41)
Uncontrolled	56	27	15	21	119	–
Total	79	51	31	52	213	–
Usual care						
Controlled	15	23	15	29	82	–
Uncontrolled	61	35	16	18	130	–
Total	76	58	31	47	212	–
Site 5						
Combined risk factor control (HbA_{1c} level <7% plus BP <130/80 mm Hg and/or LDLc level <100 mg/dL [<70 mg/dL for patients with a history of CVD])						
Intervention						
Controlled	0	24	11	33	68	8.02 (3.11–20.62)
Uncontrolled	80	56	20	43	199	–
Total	80	80	31	76	267	–
Usual care						
Controlled	0	2	3	3	8	–
Uncontrolled	80	77	24	74	255	–
Total	80	79	27	77	263	–
HbA_{1c} level <7%						
Intervention						
Controlled	0	30	13	34	77	8.49 (3.25–22.19)
Uncontrolled	80	50	18	43	191	–
Total	80	80	31	77	268	–
Usual care						
Controlled	0	3	3	3	9	–
Uncontrolled	80	76	24	74	254	–
Total	80	79	27	77	263	–
BP <130/80 mm Hg						
Intervention						
Controlled	16	53	25	68	162	2.35 (1.18–4.70)†
Uncontrolled	64	27	6	8	105	–
Total	80	80	31	76	267	–
Usual care						
Controlled	21	48	20	56	145	–
Uncontrolled	59	31	7	21	118	–
Total	80	79	27	77	263	–
LDLc level <100 mg/dL (<70 mg/dL for patients with a history of CVD)						
Intervention						
Controlled	17	42	24	59	142	1.45 (0.79–2.67)†
Uncontrolled	63	38	7	18	126	–
Total	80	80	31	77	268	–

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Appendix Table 3–Continued

Primary Outcome	Participants, *n*					Relative Risk (95% CI)*
	Baseline	12 mo	24 mo	End of Study	Total	
Usual care						
Controlled	12	32	19	54	117	–
Uncontrolled	68	47	8	23	146	–
Total	80	79	27	77	263	–
Site 6						
Combined risk factor control (HbA_{1c} level <7% plus BP <130/80 mm Hg and/or LDLc level <100 mg/dL [<70 mg/dL for patients with a history of CVD])						
Intervention						
Controlled	0	0	0	2	2	NA
Uncontrolled	40	29	18	29	116	–
Total	40	29	18	31	118	–
Usual care						
Controlled	0	1	2	6	9	–
Uncontrolled	41	34	16	30	121	–
Total	41	35	18	36	130	–
HbA_{1c} level <7%						
Intervention						
Controlled	0	1	0	3	4	NA
Uncontrolled	40	28	18	28	114	–
Total	40	29	18	31	118	–
Usual care						
Controlled	0	3	3	7	13	–
Uncontrolled	41	32	15	29	117	–
Total	41	35	18	36	130	–
BP <130/80 mm Hg						
Intervention						
Controlled	9	13	7	9	38	0.91 (0.57–1.46)
Uncontrolled	31	16	11	22	80	–
Total	40	29	18	31	118	–
Usual care						
Controlled	6	14	7	12	39	–
Uncontrolled	35	21	12	24	92	–
Total	41	35	19	36	131	–
LDLc level <100 mg/dL (<70 mg/dL for patients with a history of CVD)						
Intervention						
Controlled	12	18	8	14	52	1.05 (0.74–1.49)
Uncontrolled	28	11	10	17	66	–
Total	40	29	18	31	118	–
Usual care						
Controlled	12	13	9	18	52	–
Uncontrolled	29	22	11	18	80	–
Total	41	35	20	36	132	–
Site 7						
Combined risk factor control (HbA_{1c} level <7% plus BP <130/80 mm Hg and/or LDLc level <100 mg/dL [<70 mg/dL for patients with a history of CVD])						
Intervention						
Controlled	0	9	3	10	22	3.49 (1.23–9.92)
Uncontrolled	80	59	23	53	215	–
Total	80	68	26	63	237	–
Usual care						
Controlled	0	2	1	4	7	–
Uncontrolled	80	75	30	61	246	–
Total	80	77	31	65	253	–
HbA_{1c} level <7%						
Intervention						
Controlled	0	9	5	11	25	2.98 (1.19–7.44)
Uncontrolled	80	59	21	54	214	–
Total	80	68	26	65	239	–
Usual care						
Controlled	0	3	2	4	9	–
Uncontrolled	80	74	29	61	244	–
Total	80	77	31	65	253	–

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Appendix Table 3–Continued

Primary Outcome	Participants, *n*					Relative Risk (95% CI)*
	Baseline	12 mo	24 mo	End of Study	Total	
BP <130/80 mm Hg						
Intervention						
Controlled	6	46	16	43	111	2.83 (1.58–5.08)†
Uncontrolled	74	25	10	21	130	–
Total	80	71	26	64	241	–
Usual care						
Controlled	7	29	13	26	75	–
Uncontrolled	73	48	18	39	178	–
Total	80	77	31	65	253	–
LDLc level <100 mg/dL (<70 mg/dL for patients with a history of CVD)						
Intervention						
Controlled	21	34	13	40	108	1.72 (0.95–3.12)†
Uncontrolled	59	34	13	25	131	–
Total	80	68	26	65	239	–
Usual care						
Controlled	26	27	12	40	105	–
Uncontrolled	54	50	19	25	148	–
Total	80	77	31	65	253	–
Site 8						
Combined risk factor control (HbA_{1c} level <7% plus BP <130/80 mm Hg and/or LDLc level <100 mg/dL [<70 mg/dL for patients with a history of CVD])						
Intervention						
Controlled	0	12	8	20	40	2.11 (1.12–3.97)
Uncontrolled	70	50	19	39	178	–
Total	70	62	27	59	218	–
Usual care						
Controlled	0	8	3	6	17	–
Uncontrolled	70	53	27	52	202	–
Total	70	61	30	58	219	–
HbA_{1c} level <7%						
Intervention						
Controlled	0	13	10	23	46	1.77 (1.02–3.07)
Uncontrolled	70	49	17	36	172	–
Total	70	62	27	59	218	–
Usual care						
Controlled	0	9	6	9	24	–
Uncontrolled	70	52	24	50	196	–
Total	70	61	30	59	220	–
BP <130/80 mm Hg						
Intervention						
Controlled	3	15	9	20	47	2.30 (1.15–4.63)†
Uncontrolled	67	47	18	39	171	–
Total	70	62	27	59	218	–
Usual care						
Controlled	9	14	5	14	42	–
Uncontrolled	61	47	25	45	178	–
Total	70	61	30	59	220	–
LDLc level <100 mg/dL (<70 mg/dL for patients with a history of CVD)						
Intervention						
Controlled	30	43	24	42	139	2.05 (1.07–3.93)†
Uncontrolled	40	19	3	17	79	–
Total	70	62	27	59	218	–
Usual care						
Controlled	27	39	14	38	118	–
Uncontrolled	43	22	16	21	102	–
Total	70	61	30	59	220	–
Site 9						
Combined risk factor control (HbA_{1c} level <7% plus BP <130/80 mm Hg and/or LDLc level <100 mg/dL [<70 mg/dL for patients with a history of CVD])						
Intervention						
Controlled	0	9	3	1	13	2.97 (1.06–8.29)‡
Uncontrolled	32	23	13	31	99	–
Total	32	32	16	32	112	–

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Appendix Table 3–Continued

Primary Outcome	Participants, *n*					Relative Risk (95% CI)*
	Baseline	12 mo	24 mo	End of Study	Total	
Usual care						
Controlled	0	2	0	2	4	–
Uncontrolled	31	28	11	27	97	–
Total	31	30	11	29	101	–
HbA_{1c} level <7%						
Intervention						
Controlled	0	10	3	3	16	2.14 (0.94–4.89)‡
Uncontrolled	32	22	13	29	96	–
Total	32	32	16	32	112	–
Usual care						
Controlled	0	3	0	4	7	–
Uncontrolled	31	28	11	25	95	–
Total	31	31	11	29	102	–
BP <130/80 mm Hg						
Intervention						
Controlled	5	8	5	15	33	2.37 (1.06–5.29)
Uncontrolled	27	24	11	17	79	–
Total	32	32	16	32	112	–
Usual care						
Controlled	5	3	1	7	16	–
Uncontrolled	26	28	10	22	86	–
Total	31	31	11	29	102	–
LDLc level <100 mg/dL (<70 mg/dL for patients with a history of CVD)						
Intervention						
Controlled	9	19	11	22	61	1.59 (0.98–2.58)
Uncontrolled	23	13	5	10	51	–
Total	32	32	16	32	112	–
Usual care						
Controlled	8	14	3	10	35	–
Uncontrolled	23	16	8	19	66	–
Total	31	30	11	29	101	–
Site 10						
Combined risk factor control (HbA_{1c} level <7% plus BP <130/80 mm Hg and/or LDLc level <100 mg/dL [<70 mg/dL for patients with a history of CVD])						
Intervention						
Controlled	0	8	5	6	19	2.32 (0.98–5.50)
Uncontrolled	33	24	2	25	84	–
Total	33	32	7	31	103	–
Usual care						
Controlled	0	6	2	2	10	–
Uncontrolled	33	27	6	31	97	–
Total	33	33	8	33	107	–
HbA_{1c} level <7%						
Intervention						
Controlled	0	10	5	7	22	2.27 (1.07–4.84)
Uncontrolled	33	22	2	24	81	–
Total	33	32	7	31	103	–
Usual care						
Controlled	0	6	2	3	11	–
Uncontrolled	33	27	6	30	96	–
Total	33	33	8	33	107	–
BP <130/80 mm Hg						
Intervention						
Controlled	8	16	5	20	49	1.11 (0.92–1.35)
Uncontrolled	25	16	2	11	54	–
Total	33	32	7	31	103	–
Usual care						
Controlled	11	18	5	13	47	–
Uncontrolled	22	15	3	20	60	–
Total	33	33	8	33	107	–
LDLc level <100 mg/dL (<70 mg/dL for patients with a history of CVD)						
Intervention						
Controlled	5	20	6	22	53	2.21 (0.81–6.08)
Uncontrolled	28	12	1	9	50	–
Total	33	32	7	31	103	–

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Appendix Table 3–Continued

Primary Outcome	Participants, *n*					Relative Risk (95% CI)*
	Baseline	12 mo	24 mo	End of Study	Total	
Usual care						
Controlled	7	19	4	20	50	–
Uncontrolled	26	14	4	13	57	–
Total	33	33	8	33	107	–

BP = blood pressure; CVD = cardiovascular disease; HbA_{1c} = glycated hemoglobin; LDLc = low-density lipoprotein cholesterol; NA = not applicable (could not obtain estimates).
* Overall relative risk was obtained via log-binomial models using generalized estimating equations (where the log-binomial model did not converge, we used logistic models for the estimates and report odds ratios [95% CIs]). Estimates combine all nonmissing values collected at baseline, 12 and 24 mo, and the end of the study. Model terms include treatment, time, treatment-by-time interaction, baseline value, and site.
† Odds ratio (95% CI).
‡ Relative risk for the model that included baseline value, treatment, and time (estimates were not estimable with the interaction term).

Appendix Table 4. Serious Adverse Events, by Treatment Group

Number	Description	Incident Events, *n*			Total Events Reported in Trial, *n**		
		Total (*n* = 1146)	Intervention (*n* = 575)	Usual Care (*n* = 571)	Total (*n* = 1146)	Intervention (*n* = 575)	Usual Care (*n* = 571)
1	Death	26	12	14	–	–	–
2	Transient ischemic attack	6	5	1	–	–	–
3	Ischemic stroke	6	2	4	–	–	–
4	Intracerebral hemorrhage	3	3	0	–	–	–
5	Stable angina	4	1	3	–	–	–
6	Acute coronary syndrome	19	7	12	22	10	12
7	Arrhythmia	3	2	1	–	–	–
8	Gastrointestinal bleeding†	1	0	1	–	–	–
9	Diabetic retinopathy	41	20	21	60	30	30
10	Infection	35	24	11	40	28	12
11	Renal failure†	3	0	3	–	–	–
12	Revascularization/surgical procedure	106	55	51	107	56	51
13	Hospitalization	81	41	40	135	78	57
14	Neuropathy	164	75	89	264	124	140
15	Mild hypoglycemia	154	133	21	–	–	–
16	Hypoglycemia requiring hospitalization	5	4	1	–	–	–

* Includes incident and recurrent events reported during the study. When no data are shown in the total events column, the total reported events were all incident events.
† No baseline data available for assessment of incident/recurrent events.

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